

University of Bristol
Applied Financial Econometrics Coursework
ECONM0009

**Unconventional Monetary Policy and Financial Markets:
A VECM Analysis of the Eurozone**

Student ID: 2676835
University Username: ta25219

Table of Contents

1. Introduction.....	1
2. Literature Review	1
3. Data.....	2
3.1 Variables	2
3.2 Time Series Plots	3
3.3 Unit Root Tests.....	4
4. Methodology.....	6
4.1 VAR/VECM Framework	6
4.2 Lag Selection	6
4.3 Johansen Cointegration Test	6
4.4 VECM Specification	6
4.5 Impulse Response Functions	7
4.6 Diagnostic Checks	7
5. Results	8
5.1 Johansen Cointegration Test	8
5.2 Long Run Relationship	8
5.3 Short Run Dynamics	9
5.4 Impulse Response Functions	9
5.5 Diagnostic Checks	10
6. Conclusion	11
7. References	12
8. Appendix	14
8.1 R Script.....	14
8.2 Stata Do-File.....	32

1. Introduction

Negative interest rates were introduced in June 2014 by the European Central Bank, marking a departure from conventional monetary policy and challenging standard monetary theory. This study uses a trivariate VECM framework with the Euro STOXX 50 index, the USD per Euro nominal exchange rate and the 3-month EURIBOR rate to examine interactions between equity markets, exchange rates and interest rates. The sample covers January 2015 to December 2018 at daily frequency, covering the ECB's quantitative easing announcement and the period of negative EURIBOR rates. The results indicate at least one cointegrating relationship but limited adjustment toward equilibrium, suggesting monetary transmission may differ under unconventional policy conditions.

2. Literature Review

There is a rich literature exploring the relationship between stock markets, interest rates and exchange rates. Fama (1981) established a negative relationship between stock markets and interest rates via the discount rate channel. Cassola and Morana (2004) found that equity markets play a significant role in monetary transmission and that monetary policy helps stabilise stock markets in the euro area.

Bahmani-Oskooee and Sohrabian (1992) found bi-directional causality between stock markets and exchange rates. Nieh and Lee (2001) concluded that there was no long-run relationship between the two in G7 countries, while Phylaktis and Ravazzolo (2005) found cointegration via the goods market channel in Pacific Basin countries.

Some studies have also considered all three variables. Mok (1993) reported sporadic unidirectional and weak bi-directional causality between stock prices and interest and exchange rates respectively. Maysami and Koh (2000) employ a VECM, finding that interest rates and exchange rates contribute significantly to a cointegrating relationship.

Most prior studies rely on monthly or annual data, focus on Asian or emerging markets, and do not consider the unprecedented negative interest rate environment. This study employs a VECM framework to the Eurozone, examining the joint dynamics of stock markets, interest rates and exchange rates using daily data during a period of unconventional monetary policy.

3. Data

3.1 Variables

This analysis uses three financial variables: the Euro STOXX 50 index (HDK), the USD per Euro nominal exchange rate (nom_ER), and the 3-month Euribor rate (EURIBOR). The stock index and exchange rate are sourced from Yahoo Finance via the Data Bank on Blackboard, while the interest rate is sourced from the European Money Markets Institute via Kaggle (Smit, 2020). All variables are at the daily frequency, and for both HDK and nom_ER the closing price is used.

The first two variables are log-transformed to allow for interpretation in percentage terms and to stabilise variance, while the interest rate is retained in levels due to the presence of negative values.

The sample period spans January 2015 to December 2018, a time characterised by unconventional monetary policy, including negative interest rates and quantitative easing implemented by the European Central Bank. After merging the series on common trading days, the final dataset contains 994 observations.

Table 1 presents the summary statistics of the variables' first differences. Although summary statistics may not always be the most informative in time series analysis, the high skewness and kurtosis suggest non-normality of returns, a common feature of daily financial return data.

Table 1: Summary Statistics of First Differences (Returns)

	rHDK	rn timer_ER	dEURIBOR
Mean	-0.000128	-0.0000556	-0.000387
St. Dev.	0.0118	0.00582	0.00137
Min	-0.128	-0.0258	-0.012
Max	0.055	0.0281	0.011
Skewness	-1.40	0.0254	-1.53
Kurtosis	18.0	5.22	18.1

3.2 Time Series Plots

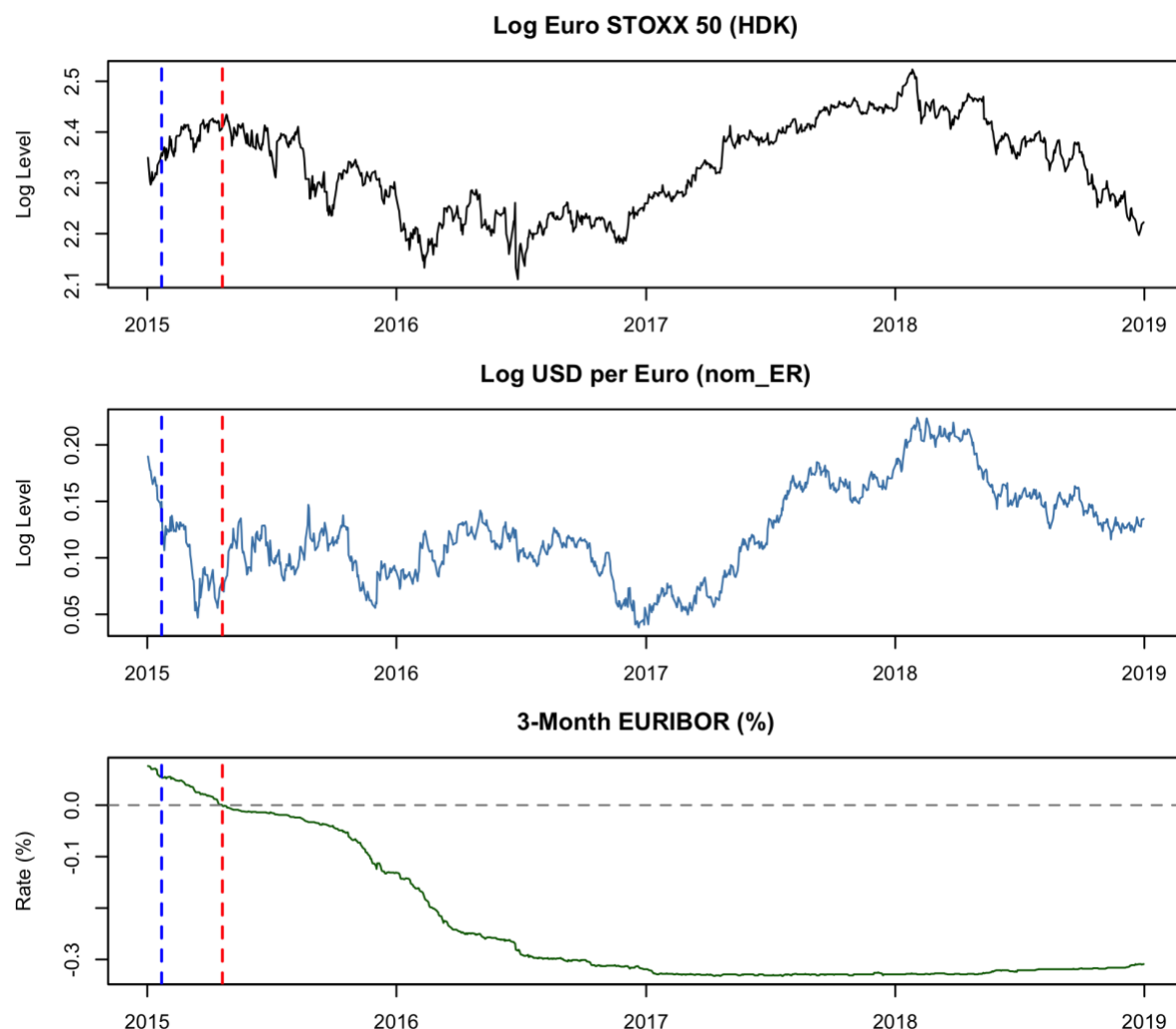


Figure 1: Log levels of the Euro STOXX 50 index, USD per Euro exchange rate and 3-month EURIBOR rate (January 2015 - December 2018). Vertical dashed lines indicate: blue = "ECB announces expanded asset purchase programme" (European Central Bank, 2015) (22nd January 2015), red = EURIBOR turns negative (21st April 2015).

Figures 1 and 2 illustrate the levels and first-differenced series of the three variables, respectively. In levels, all variables exhibit non-stationarity, with no fixed mean and persistent movements. EURIBOR displays a clear downward trend, particularly in the early part of the sample, reflecting accommodative monetary policy. The exchange rate initially stabilises following the announcement of QE in 2015 before depreciating and later appreciating as interest rates turn negative, contrary to standard macroeconomic theory. There is also some visual co-movement between the stock index and exchange rate, motivating cointegration analysis. In contrast, the first-differenced series fluctuate around zero, indicating stationarity and suggesting the variables are integrated of order one, which is tested for in the following section.

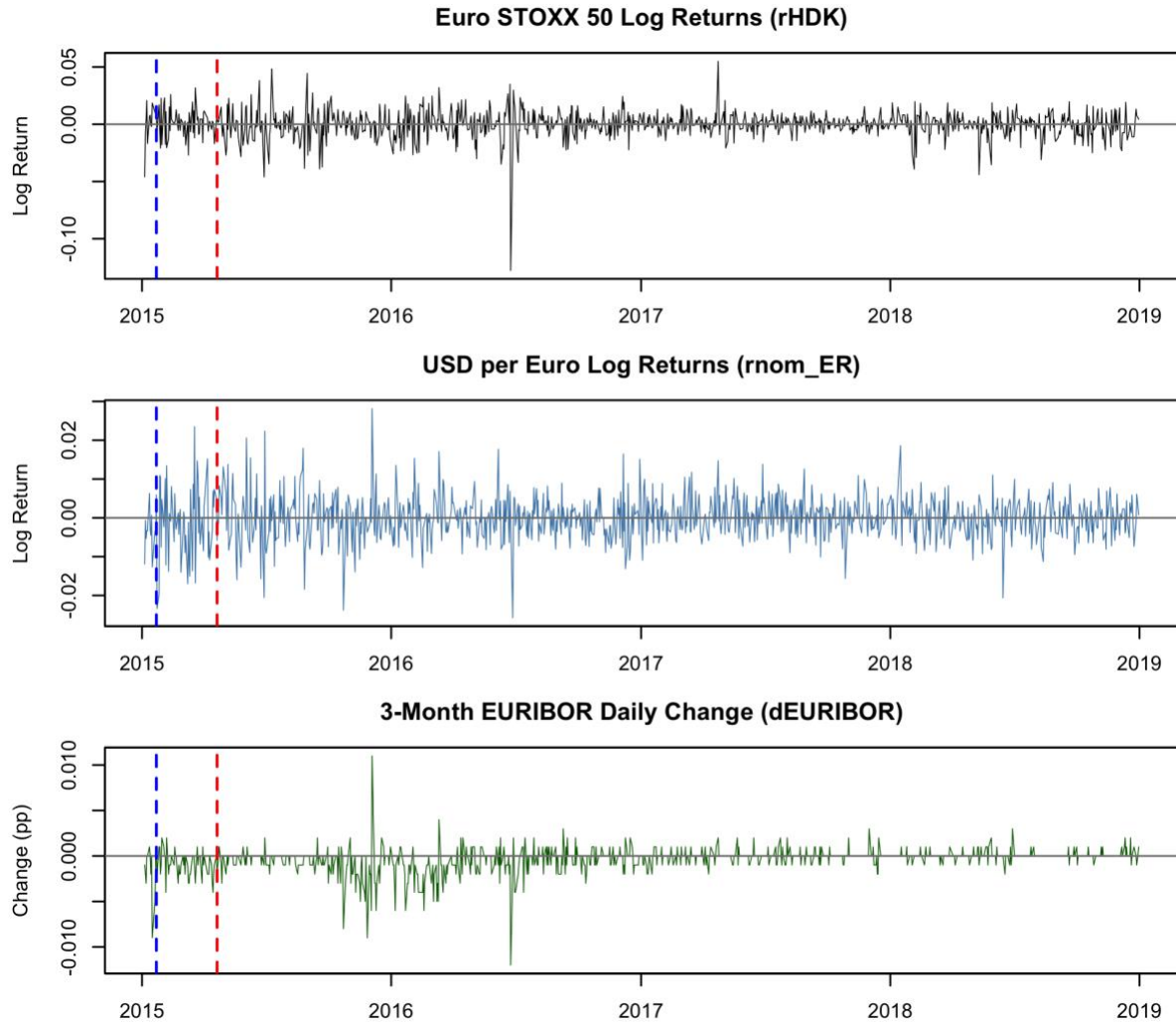


Figure 2: Log returns of the Euro STOXX 50 index, USD per Euro exchange rate and first difference of 3-month EURIBOR rate (January 2015 - December 2018). Vertical dashed lines indicate: blue = "ECB announces expanded asset purchase programme" (European Central Bank, 2015) (22nd January 2015), red = EURIBOR turns negative (21st April 2015).

3.3 Unit Root Tests

Table 2 reports the results of the Augmented Dickey-Fuller Tests (Dickey and Fuller, 1981) for all three variables. Both drift and trend specifications are reported for robustness. The maximum lag length is selected using the Schwert (1989) rule, defined as

$$p_{max} = \left\lceil 12 \times \left(\frac{T}{100} \right)^{\frac{1}{4}} \right\rceil,$$

while the optimal lag length is determined by the Akaike information criteria (AIC) and reported in the parentheses.

The stock price index and exchange rate both fail to reject the null of a unit root in levels across specifications, indicating non-stationarity, but strongly reject in first differences. This suggests both variables are integrated of order one, as expected.

Table 2: Unit root tests

Variables	Without Trend	With Trend	Without Trend	With Trend
	Levels		First Differences	
HDK_t	-1.635(4)	-1.613(4)	-17.236(3) ***	-17.233(3) ***
nom_ER_t	-2.048(1)	-2.619(1)	-23.063(1) ***	-23.053(1) **
$EURIBOR_t$	-3.255(12) **	-0.866(12)	-3.600(21) ***	-4.948(21) ***

Significance codes: *** < 0.01, ** < 0.05, * < 0.1

The results for EURIBOR are less clear, with conflicting outcomes across specifications. The unit root null is rejected at the 5% level without a trend, but not when a trend is included. Given the clear downward trend observed in Figure 1, I consider the specification including a trend more appropriate. EURIBOR is therefore classified as $I(1)$, consistent with the other variables, although this ambiguity is acknowledged as a limitation. Overall, all variables are found to be $I(1)$, satisfying a necessary condition for cointegration and motivating the Johansen test.

4. Methodology

4.1 VAR/VECM Framework

Since all three variables are $I(1)$, a multivariate time series framework is suitable in this context. A Vector Autoregressive (VAR) model is able to capture any dynamic interactions between variables, without imposing any exogeneity assumptions on variables. However, in the presence of cointegration, a VAR model would be misspecified, motivating the use of a Vector Error Correction Model (VECM) instead. Engle and Granger (1987) proposed the theoretical basis of the Error Correction Model, while Johansen (1991) provided the multivariate extension. The VECM incorporates an error correction term which captures the adjustment toward a long-run equilibrium, while simultaneously capturing short-run dynamics.

4.2 Lag Selection

The optimal lag length is selected using `VARselect` in R. For consistency with the unit root testing, I use AIC to determine the optimal lag length and keep the maximum lag length as determined by the Schwert (1989) rule. Selecting the appropriate lag length is important to ensure residuals are white noise and to avoid serial correlation.

4.3 Johansen Cointegration Test

The Johansen (1991) trace test is used to examine cointegration among the three variables. This approach is preferred to the Engle-Granger method as it allows for multiple cointegrating relationships and treats all variables symmetrically. It tests the null hypothesis that the number of cointegrating vectors is less than or equal to r , against the alternative of greater than r , proceeding sequentially from $r = 0$.

Three specifications of deterministic components are considered. Case 4 includes a constant in the cointegrating equation only, Case 3 includes a constant in both the long- and short-run equations, while Case 2 allows for a trend in the long-run relationship. Case 3 is taken as the baseline, as the variables exhibit non-zero means and there is no strong theoretical justification for a deterministic trend.

All specifications are reported for robustness. However, as noted by Otero and Smith (2000), the relatively short time span of daily data may reduce the power of the cointegration tests.

4.4 VECM Specification

If there is evidence of cointegration, a VECM is estimated using the `cajorls` function in R. As suggested by Johansen (1991), it uses the maximum likelihood approach. The general equation of a VECM(p) is:

$$\Delta \mathbf{y}_t = \alpha(\beta' \mathbf{y}_{t-1} + \mu + \rho t) + \sum_{i=1}^{p-1} \Gamma_i \Delta \mathbf{y}_{t-i} + \gamma + \tau t + \epsilon_t$$

where,

- $\Delta \mathbf{y}_t$ is the vector of first differences of the three variables

- α is the loading matrix which captures the speed of adjustment
- $\beta' y_{t-1}$ captures the error correction term
- Γ_i are the short-run dynamic coefficients on the lagged differences
- ϵ_t is an error term
- In Case 3 mentioned earlier, ρ and τ are assumed to be zero

β defines the long-run cointegrating relationship while α captures how quickly each variable adjusts back to equilibrium. The coefficient on the error correction term (the corresponding element of the loading matrix, α) should be negative, less than one in absolute value and statistically significant for valid error correction behaviour.

4.5 Impulse Response Functions

Impulse Response Functions (IRFs) are derived from the VAR representation of VECM using the `vec2var` function in R. IRFs illustrate the dynamic response of each variable to a one standard deviation shock of another variable across a specified time horizon. I use 20 trading days, approximately equal to one trading month, as this should capture the short to medium term dynamics and is a common choice for daily financial data. Bootstrap confidence intervals are constructed since they are more reliable than asymptotic intervals for finite samples. IRFs are able to capture short-run dynamic interactions between the variables which may not be observed in the loading matrix of the VECM.

4.6 Diagnostic Checks

Following estimation, diagnostic checks are performed on the VECM residuals, including tests for serial correlation, normality and ARCH effects. I also apply the Robinson (1995) semiparametric test to the cointegrating residuals using the `roblpr` command in Stata to test for long memory. The do-file is provided in the appendix.

5. Results

5.1 Johansen Cointegration Test

Table 3 reports the results of the Johansen trace test across the three cases mentioned in Section 4.3. There is evidence of at least one cointegrating relationship at the 1% significance level, which is robust across all specifications. Case 3 suggests that there are potentially two cointegrating relationships, but only at the 10% significance level, while Cases 2 and 4 conclude there is only one. I therefore proceed with $r = 1$, given inconclusive evidence that $r = 2$. Despite the potentially reduced power of the Johansen test due to the short 4-year daily sample as suggested by Otero and Smith (2000), cointegration is still detected at the 1% significance level across all specifications. This confirms cointegration, motivating VECM estimation.

Table 3: Johansen (1991) trace test for number of cointegrating relations

Null Hypothesis	Test Statistic	10%	5%	1%	Decision
Case 4					
$r \leq 2$	2.98	7.52	9.24	12.97	Cannot reject at 10% level
$r \leq 1$	17.65	17.85	19.96	24.60	Cannot reject at 10% level
$r = 0$	110.64	32.00	34.91	41.07	Reject at 1% level
Case 3 (baseline)					
$r \leq 2$	2.34	6.50	8.18	11.65	Cannot reject at 10% level
$r \leq 1$	16.15	15.66	17.95	23.52	Reject at 10% level
$r = 0$	78.58	28.71	31.52	37.22	Reject at 1% level
Case 2					
$r \leq 2$	2.65	10.49	12.25	16.26	Cannot reject at 10% level
$r \leq 1$	16.47	22.76	25.32	30.45	Cannot reject at 10% level
$r = 0$	82.64	39.06	42.44	48.45	Reject at 1% level

5.2 Long Run Relationship

Table 4 presents the normalised cointegrating vector from the Case 3 baseline. The vector is normalised to HDK, so the coefficients represent the long-run relationship with respect to the stock index.

Table 4: Long run Cointegrating Vectors

Variable	ect1
HDK	1
nom_ER	−0.405
EURIBOR	−0.827

The cointegrating vector can be rearranged as:

$$\text{HDK} = 0.405 \times \text{nom_ER} + 0.827 \times \text{EURIBOR}$$

This equation implies that on average, a 1% appreciation in the Euro is associated with a 0.405% increase in the Euro STOXX 50 closing price in the long run. The positive relationship suggests that when the Euro strengthens, it attracts foreign investors into Eurozone markets, which pushes stock prices up.

The positive coefficient on EURIBOR contradicts standard monetary theory, which predicts that higher interest rates should reduce stock prices. One possible explanation is that given the unprecedented negative interest rate period, increases in rates may signal improving economic conditions, which can boost investor confidence and support stock prices.

5.3 Short Run Dynamics

Table 5 presents the VECM estimation results. The HDK equation has the only negative error correction term (ECT). The ECT coefficient of -0.0038 implies that 0.38% of any deviation from the long-run equilibrium is corrected each trading day, corresponding to approximately 263 trading days (roughly one year) to fully return to equilibrium. Despite being negative and less than one in absolute value, it is not statistically significant, suggesting the Euro STOXX 50 index does not adjust significantly toward the long-run equilibrium.

With exchange rates as the response, the ECT is significant at the 10% level and positive, which indicates divergence from rather than adjustment toward equilibrium, inconsistent with conventional error correction. The first lag of stock market returns is significant, suggesting stock returns explain exchange rate changes in the short run.

For EURIBOR as the response, the ECT is less than one in absolute value and statistically significant, but positive. Its own lags are significant, suggesting persistence in the short run.

Overall, none of the variables fully satisfy the conditions for valid error correction, suggesting conventional adjustment mechanisms may be disrupted during this unprecedented monetary policy period. The extremely small error correction term may also be potential evidence for a long memory time series (which is tested for in the Diagnostic Checks).

Table 5: VECM Estimation Short Run Results

	Response Variable		
	HDK	nom_ER	EURIBOR
ect1	-0.00380 (0.222)	0.00287 * (0.055)	0.00263 *** (< 0.001)
ΔHDK_{t-1}	-0.0159 (0.618)	0.120 *** (< 0.001)	-0.00839 ** (0.017)
ΔHDK_{t-2}	-0.00934 (0.776)	-0.0210 (0.184)	-0.00166 (0.644)
$\Delta\text{EURIBOR}_{t-1}$	0.474 (0.107)	-0.128 (0.364)	0.123 *** (< 0.001)
$\Delta\text{EURIBOR}_{t-2}$	-0.152 (0.604)	-0.0333 (0.813)	0.0963 *** (0.0027)
$\Delta\text{nom_ER}_{t-1}$	-0.0380 (0.570)	-0.0375 (0.243)	-0.0121 * (0.098)
$\Delta\text{nom_ER}_{t-2}$	-0.0842 (0.194)	-0.0286 (0.358)	-0.00830 (0.242)
R^2	0.00665	0.06756	0.1914

Notes: P-values in parentheses. Significance codes: *** < 0.01, ** < 0.05, * < 0.1. Full results available in appendix.

5.4 Impulse Response Functions

Figure 3 illustrates the impulse response functions from the estimated VECM. The response of the Euro STOXX 50 to both exchange rates and interest rates are mainly small, short-lived and statistically insignificant. Interestingly, the exchange rate displays a

positive, more persistent and significant response to a stock market shock. This supports the short-run dynamics, suggesting stock market movements lead exchange rates.

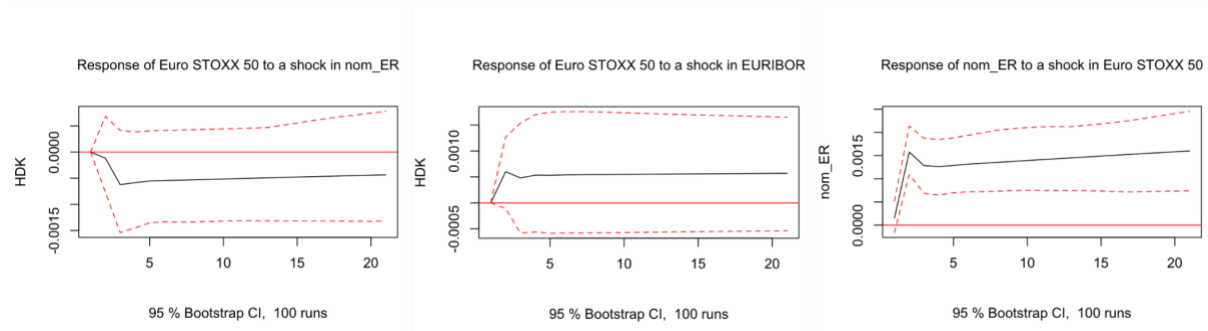


Figure 3: Impulse Response Functions. Left: Response of HDK to a shock in nom_ER; Middle: Response of HDK to a shock in EURIBOR; Right: Response of nom_ER to a shock in HDK.

5.5 Diagnostic Checks

Diagnostic checks reveal evidence of serial correlation at the optimal lag of $k = 3$. This persists across all lag orders from $k = 4$ to $k = 16$. Beyond $k = 16$, the model becomes over-parameterised. This is common in daily financial data and so is not considered model misspecification, but rather as a limitation of the analysis.

Non-normality in the residuals and ARCH effects are also detected, which are consistent with the high kurtosis seen in Table 1 and the volatility clustering visible in Figure 2. These features are also common in daily financial data and are acknowledged as limitations.

Finally, the Robinson (1995) test yields an estimated fractional differencing parameter of $d = 0.873$, so there is possible evidence of long memory cointegration here, challenging the strict $I(1)$ assumption of the Johansen test. These results should therefore be interpreted with caution.

6. Conclusion

This study finds cointegration at the 1% significance level across all specifications. A positive long-run relationship is identified between `nom_ER` and `HDK`, consistent with the capital flows channel, while the positive relationship between `EURIBOR` and `HDK` challenges standard monetary theory. In the short run, none of the variables fully satisfy the error correction conditions and equity markets appear to lead exchange rates. Impulse response functions confirm that equity shocks have a persistent positive effect on exchange rates.

However, the analysis is subject to several limitations. The sample consists of only four years of daily data, which Otero and Smith (2000) suggest may reduce the power of the Johansen test. Diagnostic checks indicate serial correlation across all lag orders, non-normality in the residuals and ARCH effects, which are common in daily financial data but mean results should be interpreted with caution. The Robinson (1995) test suggests possible fractional cointegration, which challenges the strict $I(1)$ assumption of the Johansen test. The absence of valid error correction suggests potential omitted variables. In addition, the VECM assumes parameter stability, so structural breaks associated with ECB policy changes may affect the results.

Future research could extend the sample period or consider lower-frequency data. Additional variables, such as inflation or long-term interest rates, could also be incorporated. Results could also be compared across Eurozone countries to account for potential heterogeneity. The potential fractional cointegration finding also motivates the use of fractionally cointegrated VAR models to account for long memory.

7. References

- Bahmani-Oskooee, M. and Sohrabian, A. (1992). Stock prices and the effective exchange rate of the dollar. *Applied Economics*, 24(4), pp.459–464.
doi:<https://doi.org/10.1080/00036849200000020>.
- Cassola, N. and Morana, C. (2004). Monetary policy and the stock market in the euro area. *Journal of Policy Modeling*, 26(3), pp.387–399.
doi:<https://doi.org/10.1016/j.jpolmod.2004.03.012>.
- Dickey, D.A. and Fuller, W.A. (1981). Likelihood Ratio Statistics for Autoregressive Time Series with a Unit Root. *Econometrica*, 49(4), pp.1057–1072.
doi:<https://doi.org/10.2307/1912517>.
- Engle, R.F. and Granger, C.W.J. (1987). Co-Integration and Error Correction: Representation, Estimation, and Testing. *Econometrica*, 55(2), pp.251–276.
- European Central Bank (2015). *ECB announces expanded asset purchase programme*. [online] European Central Bank. Available at:
https://www.ecb.europa.eu/press/pr/date/2015/html/pr150122_1.en.html.
- Fama, E.F. (1981). Stock Returns, Real Activity, Inflation, and Money. *The American Economic Review*, [online] 71(4), pp.545–565. doi:<https://doi.org/10.2307/1806180>.
- Johansen, S. (1991). Estimation and Hypothesis Testing of Cointegration Vectors in Gaussian Vector Autoregressive Models. *Econometrica*, 59(6), pp.1551–1580.
doi:<https://doi.org/10.2307/2938278>.
- Maysami, R.C. and Koh, T.S. (2000). A vector error correction model of the Singapore stock market. *International Review of Economics & Finance*, 9(1), pp.79–96.
doi:[https://doi.org/10.1016/s1059-0560\(99\)00042-8](https://doi.org/10.1016/s1059-0560(99)00042-8).
- Mok, H.M. (1993). Causality of interest rate, exchange rate and stock prices at stock market open and close in Hong Kong. *Asia Pacific Journal of Management*, 10(2), pp.123–143. doi:<https://doi.org/10.1007/bf01734274>.
- Nieh, C.-C. and Lee, C.-F. (2001). Dynamic relationship between stock prices and exchange rates for G-7 countries. *The Quarterly Review of Economics and Finance*, 41(4), pp.477–490. doi:[https://doi.org/10.1016/s1062-9769\(01\)00085-0](https://doi.org/10.1016/s1062-9769(01)00085-0).
- Otero, J. and Smith, J. (2000). Testing for cointegration: power versus frequency of observation — further Monte Carlo results. *Economics Letters*, 67(1), pp.5–9.
doi:[https://doi.org/10.1016/s0165-1765\(99\)00245-1](https://doi.org/10.1016/s0165-1765(99)00245-1).
- Phylaktis, K. and Ravazzolo, F. (2005). Stock prices and exchange rate dynamics. *Journal of International Money and Finance*, [online] 24(7), pp.1031–1053.
doi:<https://doi.org/10.1016/j.jimonfin.2005.08.001>.

Robinson, P.M. (1995). Gaussian Semiparametric Estimation of Long Range Dependence. *The Annals of Statistics*, [online] 23(5), pp.1630–1661. doi:<https://doi.org/10.2307/2242539>.

Schwert, G.W. (1989). Tests for Unit Roots: A Monte Carlo Investigation. *Journal of Business & Economic Statistics*, 7(2), p.147. doi:<https://doi.org/10.2307/1391432>.

Smit, J.J.H. (2020). *EURIBOR Daily Rates* [Dataset]. Kaggle. Available at: <https://www.kaggle.com/datasets/jorijnsmit/euribor-daily-rates> [Accessed: March 2026].

8. Appendix

8.1 R Script

```
## =====  
## ===== APPLIED FINANCIAL ECONOMETRICS COURSEWORK =====  
## =====  
  
# Clear workspace and set working directory  
rm(list = ls())  
setwd('/Users/hidde/Library/CloudStorage/OneDrive-UniversityofBristol/Documents/Applied Financial Econ  
ometrics/Coursework/')  
  
# Load required packages  
library(xts)  
library(urca)  
library(vars)  
library(ggplot2)  
#install.packages("moments")  
library(moments)  
  
## =====  
## SECTION 1: DATA PREPARATION  
## =====  
  
# --- 1.1 Euro STOXX 50 ---  
df <- read.csv("Euro STOXX 50 Index.csv")  
df$date <- as.Date(df$date, format = "%Y-%m-%d")  
  
# Create xts object - variable named with initials (HDK) as required  
HDK <- xts(df$Close, order.by = df$date)  
HDK <- na.omit(HDK)  
  
# --- 1.2 nom_ER Nominal Exchange Rate ---  
nom_ER <- read.csv("ER_USDollar_per_Euro.csv")  
nom_ER$date <- as.Date(nom_ER$date, format = "%Y-%m-%d")  
nom_ER$Close <- as.numeric(nom_ER$Close)  
## Warning: NAs introduced by coercion  
  
nom_ER <- xts(nom_ER$Close, order.by = nom_ER$date)  
nom_ER <- na.omit(nom_ER)  
colnames(nom_ER) <- "nom_ER"  
  
# --- 1.3 EURIBOR 3-Month Interest Rate ---  
euribor <- read.csv("euribor_daily_rates.csv")  
euribor$date <- as.Date(euribor$date, format = "%Y-%m-%d")  
  
# Extract 3-month maturity  
EURIBOR <- xts(euribor$X3m, order.by = euribor$date)  
EURIBOR <- na.omit(EURIBOR)  
colnames(EURIBOR) <- "EURIBOR"  
  
# --- 1.4 Restrict all series to January 2015 - December 2018 ---  
start_date <- as.Date("2015-01-01")  
end_date <- as.Date("2018-12-31")  
  
HDK <- HDK[index(HDK) >= start_date & index(HDK) <= end_date]  
nom_ER <- nom_ER[index(nom_ER) >= start_date & index(nom_ER) <= end_date]  
EURIBOR <- EURIBOR[index(EURIBOR) >= start_date & index(EURIBOR) <= end_date]  
  
# --- 1.5 Log transformation ---  
# Note: EURIBOR is negative/near-zero so NO log transformation applied  
# nom_ER and Euro Stoxx are strictly positive so log transformation applied  
ln_HDK <- log(HDK)  
ln_nom_ER <- log(nom_ER)  
  
# --- 1.6 Merge all three series on common trading days (inner join) ---  
# xts handles irregular daily data natively - no gaps problem  
ln_merged <- merge(ln_HDK, ln_nom_ER, join = "inner")  
ln_merged <- merge(ln_merged, EURIBOR, join = "inner")
```



```

ln_merged <- na.omit(ln_merged)
colnames(ln_merged) <- c("HDK", "nom_ER", "EURIBOR")

cat("Sample period:", format(start(ln_merged)), "to", format(end(ln_merged)), "\n")

## Sample period: 2015-01-02 to 2018-12-31

cat("Number of observations:", nrow(ln_merged), "\n")

## Number of observations: 995

# --- 1.7 Compute returns / first differences ---
# Log returns for Euro Stoxx and nom_ER
# Simple first difference for EURIBOR (percentage point change)
rHDK <- diff(ln_merged$HDK, differences = 1)
rnom_ER <- diff(ln_merged$nom_ER, differences = 1)
dEURIBOR <- diff(ln_merged$EURIBOR, differences = 1)

returns <- na.omit(merge(rHDK, rnom_ER, dEURIBOR))
colnames(returns) <- c("rHDK", "rnom_ER", "dEURIBOR")

cat("Number of return observations:", nrow(returns), "\n")

## Number of return observations: 994

# --- 1.8 Summary statistics ---
library(moments)

cat("\n--- Summary statistics: returns / first differences ---\n")

##
## --- Summary statistics: returns / first differences ---

summary_returns <- sapply(as.data.frame(returns),
                          function(x) signif(c(Mean = mean(x, na.rm = TRUE),
                                                  SD = sd(x, na.rm = TRUE),
                                                  Min = min(x, na.rm = TRUE),
                                                  Max = max(x, na.rm = TRUE),
                                                  Skew = skewness(x, na.rm = TRUE),
                                                  Kurt = kurtosis(x, na.rm = TRUE)), 3))

print(summary_returns)

##           rHDK    rnom_ER dEURIBOR
## Mean -0.000128 -5.56e-05 -0.000387
## SD    0.011800  5.82e-03  0.001370
## Min   -0.128000 -2.58e-02 -0.012000
## Max    0.055000  2.81e-02  0.011000
## Skew  -1.400000  2.54e-02 -1.530000
## Kurt  18.000000  5.22e+00 18.100000

## =====
## SECTION 2: PLOTS
## =====

# Key event dates for 2015-2018 sample
qe_start <- as.Date("2015-01-22") # ECB QE announcement
euribor_neg <- as.Date("2015-04-21") # EURIBOR 3m turns negative

# --- Plot A: Log Levels ---
png("plot_levels.png", width = 1950, height = 1700, res = 300)
par(mfrow = c(3, 1), mar = c(2, 4, 3, 1))

# Panel 1: Euro STOXX 50 Log Levels
plot(as.zoo(ln_merged[, "HDK"]),
     main = "Log Euro STOXX 50 (HDK)",
     ylab = "Log Level",
     xlab = "",
     col = "black",
     lwd = 1)
abline(v = qe_start, col = "blue", lty = 2, lwd = 1.5)
abline(v = euribor_neg, col = "red", lty = 2, lwd = 1.5)

# Panel 2: nom_ER Log Levels

```

```

plot(as.zoo(ln_merged[, "nom_ER"]),
     main = "Log USD per Euro (nom_ER)",
     ylab = "Log Level",
     xlab = "",
     col = "steelblue",
     lwd = 1)
abline(v = qe_start, col = "blue", lty = 2, lwd = 1.5)
abline(v = euribor_neg, col = "red", lty = 2, lwd = 1.5)

# Panel 3: EURIBOR Levels
plot(as.zoo(ln_merged[, "EURIBOR"]),
     main = "3-Month EURIBOR (%)",
     ylab = "Rate (%)",
     xlab = "",
     col = "darkgreen",
     lwd = 1)
abline(h = 0, col = "grey50", lty = 2, lwd = 1)
abline(v = qe_start, col = "blue", lty = 2, lwd = 1.5)
abline(v = euribor_neg, col = "red", lty = 2, lwd = 1.5)

dev.off()

## quartz_off_screen
## 2

par(mfrow = c(1, 1))

# --- Plot B: First differences ---
png("plot_returns.png", width = 1950, height = 1700, res = 300)
par(mfrow = c(3, 1), mar = c(2, 4, 3, 1))

# Panel 1: Euro STOXX 50 Log returns
plot(as.zoo(returns[, "rHDK"]),
     main = "Euro STOXX 50 Log Returns (rHDK)",
     ylab = "Log Return",
     xlab = "",
     col = "black",
     lwd = 0.5)
abline(h = 0, col = "grey50", lty = 1, lwd = 1)
abline(v = qe_start, col = "blue", lty = 2, lwd = 1.5)
abline(v = euribor_neg, col = "red", lty = 2, lwd = 1.5)

# Panel 2: nom_ER Log returns
plot(as.zoo(returns[, "rnom_ER"]),
     main = "USD per Euro Log Returns (rnom_ER)",
     ylab = "Log Return",
     xlab = "",
     col = "steelblue",
     lwd = 0.5)
abline(h = 0, col = "grey50", lty = 1, lwd = 1)
abline(v = qe_start, col = "blue", lty = 2, lwd = 1.5)
abline(v = euribor_neg, col = "red", lty = 2, lwd = 1.5)

# Panel 3: EURIBOR first differences
plot(as.zoo(returns[, "dEURIBOR"]),
     main = "3-Month EURIBOR Daily Change (dEURIBOR)",
     ylab = "Change (pp)",
     xlab = "",
     col = "darkgreen",
     lwd = 0.5)
abline(h = 0, col = "grey50", lty = 1, lwd = 1)
abline(v = qe_start, col = "blue", lty = 2, lwd = 1.5)
abline(v = euribor_neg, col = "red", lty = 2, lwd = 1.5)

dev.off()

## quartz_off_screen
## 2

par(mfrow = c(1, 1))

## =====
## SECTION 3: UNIT ROOT TESTS (ADF)
## =====

```

```

# Maximum Lag Length selected using Schwert (1989) formula: floor(12*(T/100)^0.25)
# Optimal Lag within this maximum selected by AIC
T <- nrow(ln_merged)
lag_ADF <- floor(12 * (T / 100)^0.25)
cat("Sample size T =", T, "
")

## Sample size T = 995

cat("ADF max lag (Schwert 1989: floor(12*(T/100)^0.25)) =", lag_ADF, "
")

## ADF max lag (Schwert 1989: floor(12*(T/100)^0.25)) = 21

# Both drift and trend specifications reported for robustness
# Note: EURIBOR tested in Levels (not Log) as no Log transformation applied

# --- 3.1 Euro STOXX 50 (Log Levels) ---
HDK_ADF_levels_drift <- ur.df(na.omit(ln_merged$HDK),
                             type = "drift",
                             lags = lag_ADF,
                             selectlags = "AIC")
summary(HDK_ADF_levels_drift)

##
## #####
## # Augmented Dickey-Fuller Test Unit Root Test #
## #####
##
## Test regression drift
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.125637 -0.004998  0.000757  0.006046  0.055527
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.015826   0.009805   1.614   0.1068
## z.lag.1      -0.006866   0.004200  -1.635   0.1024
## z.diff.lag1  -0.004758   0.032030  -0.149   0.8819
## z.diff.lag2  -0.018055   0.031990  -0.564   0.5726
## z.diff.lag3  -0.012769   0.031974  -0.399   0.6897
## z.diff.lag4  -0.078293   0.031919  -2.453   0.0143 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.01163 on 967 degrees of freedom
## Multiple R-squared:  0.01024, Adjusted R-squared:  0.005122
## F-statistic: 2.001 on 5 and 967 DF, p-value: 0.07613
##
##
## Value of test-statistic is: -1.6349 1.4697
##
## Critical values for test statistics:
##      1pct  5pct 10pct
## tau2 -3.43 -2.86 -2.57
## phi1  6.43  4.59  3.78

HDK_ADF_levels_trend <- ur.df(na.omit(ln_merged$HDK),
                              type = "trend",
                              lags = lag_ADF,
                              selectlags = "AIC")
summary(HDK_ADF_levels_trend)

##
## #####
## # Augmented Dickey-Fuller Test Unit Root Test #
## #####
##

```

```

## Test regression trend
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + tt + z.diff.lag)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.125635 -0.005078  0.000737  0.006048  0.055509
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.620e-02  1.006e-02   1.610   0.1078
## z.lag.1      -7.075e-03  4.387e-03  -1.613   0.1072
## tt           2.296e-07  1.388e-06   0.165   0.8686
## z.diff.lag1 -4.569e-03  3.207e-02  -0.142   0.8867
## z.diff.lag2 -1.784e-02  3.203e-02  -0.557   0.5776
## z.diff.lag3 -1.256e-02  3.201e-02  -0.392   0.6948
## z.diff.lag4 -7.808e-02  3.196e-02  -2.443   0.0147 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.01164 on 966 degrees of freedom
## Multiple R-squared:  0.01027,    Adjusted R-squared:  0.004121
## F-statistic: 1.67 on 6 and 966 DF,  p-value: 0.125
##
##
## Value of test-statistic is: -1.6125 0.9879 1.3488
##
## Critical values for test statistics:
##      1pct  5pct 10pct
## tau3 -3.96 -3.41 -3.12
## phi2  6.09  4.68  4.03
## phi3  8.27  6.25  5.34

# --- 3.2 nom_ER (Log Levels) ---
nom_ER_ADF_levels_drift <- ur.df(na.omit(ln_merged$nom_ER),
                                type = "drift",
                                lags = lag_ADF,
                                selectlags = "AIC")
summary(nom_ER_ADF_levels_drift)

##
## #####
## # Augmented Dickey-Fuller Test Unit Root Test #
## #####
##
## Test regression drift
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.0257770 -0.0034295 -0.0000289  0.0033282  0.0275114
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.0011208  0.0005727   1.957   0.0506 .
## z.lag.1      -0.0090655  0.0044263  -2.048   0.0408 *
## z.diff.lag   -0.0433851  0.0320752  -1.353   0.1765
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.005713 on 970 degrees of freedom
## Multiple R-squared:  0.006594,    Adjusted R-squared:  0.004546
## F-statistic: 3.219 on 2 and 970 DF,  p-value: 0.0404
##
##
## Value of test-statistic is: -2.0481 2.0987
##
## Critical values for test statistics:
##      1pct  5pct 10pct

```

```
## tau2 -3.43 -2.86 -2.57
## phi1 6.43 4.59 3.78

nom_ER_ADF_levels_trend <- ur.df(na.omit(ln_merged$nom_ER),
                                type = "trend",
                                lags = lag_ADF,
                                selectlags = "AIC")
summary(nom_ER_ADF_levels_trend)

##
## #####
## # Augmented Dickey-Fuller Test Unit Root Test #
## #####
##
## Test regression trend
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + tt + z.diff.lag)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.0255869 -0.0035789 -0.0000288  0.0032679  0.0275531
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.085e-03  5.726e-04   1.894  0.05851 .
## z.lag.1      -1.418e-02  5.416e-03  -2.619  0.00896 **
## tt           1.306e-06  7.978e-07   1.637  0.10199
## z.diff.lag   -4.105e-02  3.208e-02  -1.280  0.20100
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.005708 on 969 degrees of freedom
## Multiple R-squared:  0.009333, Adjusted R-squared:  0.006266
## F-statistic: 3.043 on 3 and 969 DF, p-value: 0.0281
##
##
## Value of test-statistic is: -2.6187 2.2946 3.4406
##
## Critical values for test statistics:
##      1pct  5pct 10pct
## tau3 -3.96 -3.41 -3.12
## phi2 6.09 4.68 4.03
## phi3 8.27 6.25 5.34

# --- 3.3 EURIBOR (levels) ---
EURIBOR_ADF_levels_drift <- ur.df(na.omit(ln_merged$EURIBOR),
                                type = "drift",
                                lags = lag_ADF,
                                selectlags = "AIC")
summary(EURIBOR_ADF_levels_drift)

##
## #####
## # Augmented Dickey-Fuller Test Unit Root Test #
## #####
##
## Test regression drift
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.0111065 -0.0003140  0.0000382  0.0005275  0.0127504
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.0004105  0.0001075  -3.821 0.000142 ***
## z.lag.1      -0.0011600  0.0003563  -3.255 0.001172 **
## z.diff.lag1   0.0771195  0.0320320   2.408 0.016247 *
## z.diff.lag2   0.0782504  0.0320336   2.443 0.014755 *
```

```

## z.diff.lag3 0.0211617 0.0317942 0.666 0.505837
## z.diff.lag4 0.1011259 0.0317456 3.186 0.001492 **
## z.diff.lag5 -0.0096349 0.0318769 -0.302 0.762525
## z.diff.lag6 0.0223465 0.0318379 0.702 0.482922
## z.diff.lag7 0.0216971 0.0318443 0.681 0.495815
## z.diff.lag8 -0.0281446 0.0318422 -0.884 0.376983
## z.diff.lag9 0.0793934 0.0316467 2.509 0.012280 *
## z.diff.lag10 0.1366479 0.0317142 4.309 1.81e-05 ***
## z.diff.lag11 0.0322159 0.0318004 1.013 0.311285
## z.diff.lag12 0.1082826 0.0309678 3.497 0.000493 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.001227 on 959 degrees of freedom
## Multiple R-squared: 0.1669, Adjusted R-squared: 0.1556
## F-statistic: 14.78 on 13 and 959 DF, p-value: < 2.2e-16
##
##
## Value of test-statistic is: -3.2554 7.4222
##
## Critical values for test statistics:
##      1pct 5pct 10pct
## tau2 -3.43 -2.86 -2.57
## phi1 6.43 4.59 3.78

EURIBOR_ADF_levels_trend <- ur.df(na.omit(ln_merged$EURIBOR),
                                type = "trend",
                                lags = lag_ADF,
                                selectlags = "AIC")

summary(EURIBOR_ADF_levels_trend)

##
## #####
## # Augmented Dickey-Fuller Test Unit Root Test #
## #####
##
## Test regression trend
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + tt + z.diff.lag)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.0110428 -0.0003373  0.0000045  0.0005558  0.0126652
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -4.785e-04  1.169e-04  -4.094 4.59e-05 ***
## z.lag.1      -4.969e-04  5.737e-04  -0.866 0.386668
## tt           4.135e-07  2.805e-07   1.474 0.140806
## z.diff.lag1  7.380e-02  3.209e-02   2.300 0.021690 *
## z.diff.lag2  7.482e-02  3.210e-02   2.331 0.019957 *
## z.diff.lag3  1.735e-02  3.188e-02   0.544 0.586482
## z.diff.lag4  9.722e-02  3.184e-02   3.054 0.002323 **
## z.diff.lag5 -1.318e-02  3.195e-02  -0.413 0.680010
## z.diff.lag6  1.851e-02  3.192e-02   0.580 0.562097
## z.diff.lag7  1.796e-02  3.193e-02   0.563 0.573822
## z.diff.lag8 -3.176e-02  3.192e-02  -0.995 0.320013
## z.diff.lag9  7.526e-02  3.175e-02   2.370 0.017976 *
## z.diff.lag10 1.326e-01  3.182e-02   4.166 3.37e-05 ***
## z.diff.lag11 2.832e-02  3.189e-02   0.888 0.374662
## z.diff.lag12 1.042e-01  3.107e-02   3.355 0.000824 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.001227 on 958 degrees of freedom
## Multiple R-squared: 0.1688, Adjusted R-squared: 0.1566
## F-statistic: 13.89 on 14 and 958 DF, p-value: < 2.2e-16
##
##
## Value of test-statistic is: -0.8661 5.6784 6.3918
##
## Critical values for test statistics:

```

```

##      1pct  5pct 10pct
## tau3 -3.96 -3.41 -3.12
## phi2  6.09  4.68  4.03
## phi3  8.27  6.25  5.34

# --- 3.4 Euro STOXX 50 (Log returns) ---
HDK_ADF_diff_drift <- ur.df(na.omit(returns$rHDK),
                           type      = "drift",
                           lags      = lag_ADF,
                           selectlags = "AIC")
summary(HDK_ADF_diff_drift)

##
## #####
## # Augmented Dickey-Fuller Test Unit Root Test #
## #####
##
## Test regression drift
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.125023 -0.005114  0.000720  0.005713  0.055551
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.0001760  0.0003735  -0.471   0.6375
## z.lag.1      -1.1235128  0.0651832 -17.236 <2e-16 ***
## z.diff.lag1   0.1175574  0.0559714   2.100   0.0360 *
## z.diff.lag2   0.0978513  0.0453090   2.160   0.0310 *
## z.diff.lag3   0.0793136  0.0319198   2.485   0.0131 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.01164 on 967 degrees of freedom
## Multiple R-squared:  0.5061, Adjusted R-squared:  0.5041
## F-statistic: 247.7 on 4 and 967 DF,  p-value: < 2.2e-16
##
##
## Value of test-statistic is: -17.2362 148.5443
##
## Critical values for test statistics:
##      1pct  5pct 10pct
## tau2 -3.43 -2.86 -2.57
## phi1  6.43  4.59  3.78

HDK_ADF_diff_trend <- ur.df(na.omit(returns$rHDK),
                           type      = "trend",
                           lags      = lag_ADF,
                           selectlags = "AIC")
summary(HDK_ADF_diff_trend)

##
## #####
## # Augmented Dickey-Fuller Test Unit Root Test #
## #####
##
## Test regression trend
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + tt + z.diff.lag)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.125083 -0.005111  0.000597  0.005731  0.055587
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  8.262e-05  7.720e-04   0.107   0.9148
## z.lag.1      -1.124e+00  6.524e-02 -17.233 <2e-16 ***

```

```

## tt          -5.099e-07  1.332e-06  -0.383   0.7019
## z.diff.lag1  1.181e-01  5.602e-02  2.109   0.0352 *
## z.diff.lag2  9.825e-02  4.534e-02  2.167   0.0305 *
## z.diff.lag3  7.949e-02  3.194e-02  2.489   0.0130 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.01164 on 966 degrees of freedom
## Multiple R-squared:  0.5062, Adjusted R-squared:  0.5036
## F-statistic: 198 on 5 and 966 DF, p-value: < 2.2e-16
##
##
## Value of test-statistic is: -17.2326 98.991 148.4861
##
## Critical values for test statistics:
##      1pct  5pct 10pct
## tau3 -3.96 -3.41 -3.12
## phi2  6.09  4.68  4.03
## phi3  8.27  6.25  5.34

# --- 3.5 nom_ER (log returns) ---
nom_ER_ADF_diff_drift <- ur.df(na.omit(returns$rnom_ER),
                              type      = "drift",
                              lags      = lag_ADF,
                              selectlags = "AIC")

summary(nom_ER_ADF_diff_drift)

##
## #####
## # Augmented Dickey-Fuller Test Unit Root Test #
## #####
##
## Test regression drift
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.0256401 -0.0033750 -0.0001117  0.0032784  0.0281911
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.064e-06  1.834e-04  -0.006   0.995
## z.lag.1      -1.071e+00  4.643e-02 -23.063 <2e-16 ***
## z.diff.lag    2.104e-02  3.208e-02  0.656   0.512
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.005717 on 969 degrees of freedom
## Multiple R-squared:  0.5254, Adjusted R-squared:  0.5245
## F-statistic: 536.5 on 2 and 969 DF, p-value: < 2.2e-16
##
##
## Value of test-statistic is: -23.0628 265.9472
##
## Critical values for test statistics:
##      1pct  5pct 10pct
## tau2 -3.43 -2.86 -2.57
## phi1  6.43  4.59  3.78

nom_ER_ADF_diff_trend <- ur.df(na.omit(returns$rnom_ER),
                              type      = "trend",
                              lags      = lag_ADF,
                              selectlags = "AIC")

summary(nom_ER_ADF_diff_trend)

##
## #####
## # Augmented Dickey-Fuller Test Unit Root Test #
## #####
##
## Test regression trend

```



```
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + tt + z.diff.lag)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.0256169 -0.0033770 -0.0000948  0.0032409  0.0282373
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -8.558e-05  3.792e-04  -0.226   0.821
## z.lag.1      -1.071e+00  4.646e-02 -23.053 <2e-16 ***
## tt           1.665e-07  6.539e-07   0.255   0.799
## z.diff.lag   2.106e-02  3.209e-02   0.656   0.512
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.00572 on 968 degrees of freedom
## Multiple R-squared:  0.5255, Adjusted R-squared:  0.524
## F-statistic: 357.3 on 3 and 968 DF, p-value: < 2.2e-16
##
##
## Value of test-statistic is: -23.0526 177.1487 265.7221
##
## Critical values for test statistics:
##      1pct  5pct 10pct
## tau3 -3.96 -3.41 -3.12
## phi2  6.09  4.68  4.03
## phi3  8.27  6.25  5.34

# --- 3.6 EURIBOR (first differences) ---
EURIBOR_ADF_diff_drift <- ur.df(na.omit(returns$dEURIBOR),
                                type = "drift",
                                lags = lag_ADF,
                                selectlags = "AIC")
summary(EURIBOR_ADF_diff_drift)

##
## #####
## # Augmented Dickey-Fuller Test Unit Root Test #
## #####
##
## Test regression drift
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.0111060 -0.0004053  0.0000789  0.0004093  0.0128079
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -8.316e-05  4.640e-05  -1.792 0.073424 .
## z.lag.1      -2.297e-01  6.381e-02  -3.600 0.000335 ***
## z.diff.lag1  -6.959e-01  6.837e-02 -10.179 < 2e-16 ***
## z.diff.lag2  -6.182e-01  7.184e-02  -8.605 < 2e-16 ***
## z.diff.lag3  -6.051e-01  7.429e-02  -8.145 1.19e-15 ***
## z.diff.lag4  -5.019e-01  7.640e-02  -6.569 8.32e-11 ***
## z.diff.lag5  -5.099e-01  7.768e-02  -6.564 8.59e-11 ***
## z.diff.lag6  -4.913e-01  7.901e-02  -6.218 7.53e-10 ***
## z.diff.lag7  -4.677e-01  7.995e-02  -5.851 6.74e-09 ***
## z.diff.lag8  -4.942e-01  8.044e-02  -6.143 1.19e-09 ***
## z.diff.lag9  -4.059e-01  8.112e-02  -5.004 6.68e-07 ***
## z.diff.lag10 -2.527e-01  8.107e-02  -3.117 0.001881 **
## z.diff.lag11 -2.152e-01  7.955e-02  -2.705 0.006954 **
## z.diff.lag12 -8.633e-02  7.805e-02  -1.106 0.268998
## z.diff.lag13 -5.773e-02  7.547e-02  -0.765 0.444520
## z.diff.lag14 -5.311e-02  7.211e-02  -0.737 0.461581
## z.diff.lag15 -1.978e-02  6.875e-02  -0.288 0.773651
## z.diff.lag16  1.969e-02  6.467e-02   0.305 0.760812
## z.diff.lag17  7.954e-02  6.012e-02   1.323 0.186142
```

```

## z.diff.lag18 1.344e-01 5.558e-02 2.419 0.015752 *
## z.diff.lag19 9.465e-02 4.930e-02 1.920 0.055191 .
## z.diff.lag20 9.525e-02 4.189e-02 2.274 0.023212 *
## z.diff.lag21 9.817e-02 3.108e-02 3.158 0.001637 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.001225 on 949 degrees of freedom
## Multiple R-squared:  0.484, Adjusted R-squared:  0.472
## F-statistic: 40.46 on 22 and 949 DF, p-value: < 2.2e-16
##
## Value of test-statistic is: -3.6 6.4895
##
## Critical values for test statistics:
##      1pct 5pct 10pct
## tau2 -3.43 -2.86 -2.57
## phi1  6.43  4.59  3.78

EURIBOR_ADF_diff_trend <- ur.df(na.omit(returns$dEURIBOR),
                                type      = "trend",
                                lags      = lag_ADF,
                                selectlags = "AIC")
summary(EURIBOR_ADF_diff_trend)

##
## #####
## # Augmented Dickey-Fuller Test Unit Root Test #
## #####
##
## Test regression trend
##
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + tt + z.diff.lag)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.0110405 -0.0003539  0.0000133  0.0005231  0.0125718
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -4.645e-04  1.218e-04  -3.815 0.000145 ***
## z.lag.1       -4.093e-01  8.273e-02  -4.948 8.88e-07 ***
## tt           6.147e-07  1.816e-07   3.384 0.000743 ***
## z.diff.lag1  -5.268e-01  8.438e-02  -6.244 6.44e-10 ***
## z.diff.lag2  -4.588e-01  8.558e-02  -5.361 1.04e-07 ***
## z.diff.lag3  -4.541e-01  8.632e-02  -5.261 1.77e-07 ***
## z.diff.lag4  -3.582e-01  8.704e-02  -4.115 4.20e-05 ***
## z.diff.lag5  -3.731e-01  8.719e-02  -4.279 2.07e-05 ***
## z.diff.lag6  -3.625e-01  8.731e-02  -4.151 3.60e-05 ***
## z.diff.lag7  -3.468e-01  8.717e-02  -3.978 7.48e-05 ***
## z.diff.lag8  -3.810e-01  8.671e-02  -4.394 1.24e-05 ***
## z.diff.lag9  -3.011e-01  8.642e-02  -3.484 0.000516 ***
## z.diff.lag10 -1.554e-01  8.560e-02  -1.815 0.069770 .
## z.diff.lag11 -1.247e-01  8.351e-02  -1.493 0.135766
## z.diff.lag12 -3.179e-03  8.142e-02  -0.039 0.968867
## z.diff.lag13  1.708e-02  7.825e-02   0.218 0.827299
## z.diff.lag14  1.312e-02  7.434e-02   0.176 0.859998
## z.diff.lag15  3.791e-02  7.046e-02   0.538 0.590740
## z.diff.lag16  6.894e-02  6.595e-02   1.045 0.296087
## z.diff.lag17  1.209e-01  6.103e-02   1.981 0.047912 *
## z.diff.lag18  1.689e-01  5.620e-02   3.005 0.002729 **
## z.diff.lag19  1.219e-01  4.969e-02   2.453 0.014343 *
## z.diff.lag20  1.143e-01  4.205e-02   2.720 0.006656 **
## z.diff.lag21  1.082e-01  3.106e-02   3.483 0.000518 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.001219 on 948 degrees of freedom
## Multiple R-squared:  0.4902, Adjusted R-squared:  0.4778
## F-statistic: 39.63 on 23 and 948 DF, p-value: < 2.2e-16
##
##

```

```
## Value of test-statistic is: -4.9478 8.1915 12.2776
##
## Critical values for test statistics:
##      1pct  5pct 10pct
## tau3 -3.96 -3.41 -3.12
## phi2  6.09  4.68  4.03
## phi3  8.27  6.25  5.34

## =====
## SECTION 4: JOHANSEN COINTEGRATION TESTS
## =====

# Bind all three series into matrix for Johansen test
data_coint <- cbind(ln_merged$HDK, ln_merged$nom_ER, ln_merged$EURIBOR)
colnames(data_coint) <- c("HDK", "nom_ER", "EURIBOR")

# --- 4.1 Lag order selection ---
# type="both" includes constant and trend in the VAR
lag_select <- VARselect(data_coint, lag.max = lag_ADF, type = "both")
print(lag_select$selection)

## AIC(n)  HQ(n)  SC(n)  FPE(n)
##      3      2      2      3

optimal_lag <- lag_select$selection["AIC(n)"]
cat("Optimal lag (AIC):", optimal_lag, "\n")

## Optimal lag (AIC): 3

# --- 4.2 Johansen Trace Tests ---
# Three cases following Johansen (1991):
# Case 2: restricted trend (ecdet="trend")
# Case 3: unrestricted constant (ecdet="none") - baseline
# Case 4: restricted constant (ecdet="const")

# Case 4 - restricted constant
Jtest_trace_case4 <- ca.jo(data_coint,
                           type = "trace",
                           ecdet = "const",
                           K = optimal_lag,
                           spec = "transitory")
summary(Jtest_trace_case4)

##
## #####
## # Johansen-Procedure #
## #####
##
## Test type: trace statistic , without linear trend and constant in cointegration
##
## Eigenvalues (lambda):
## [1] 8.948671e-02 1.467565e-02 2.998588e-03 8.039692e-18
##
## Values of teststatistic and critical values of test:
##
##      test 10pct  5pct  1pct
## r <= 2 |   2.98  7.52  9.24 12.97
## r <= 1 |  17.65 17.85 19.96 24.60
## r = 0  | 110.64 32.00 34.91 41.07
##
## Eigenvectors, normalised to first column:
## (These are the cointegration relations)
##
##      HDK.l1  nom_ER.l1  EURIBOR.l1  constant
## HDK.l1      1.0000000  1.0000000  1.0000000  1.000000
## nom_ER.l1 -0.2686382 -3.7150744  0.09048794 -3.477233
## EURIBOR.l1 -0.8603312 -0.4480305  0.24965858  2.489704
## constant  -2.6181515 -1.9573336 -2.25106845 -1.660351
##
## Weights W:
## (This is the loading matrix)
##
##      HDK.l1  nom_ER.l1  EURIBOR.l1  constant
## HDK.d      -0.002242065 -2.331866e-04 -0.0062439762 -1.226754e-17
```

```
## nom_ER.d 0.001833917 5.682745e-03 -0.0001861510 2.557047e-17
## EURIBOR.d 0.002539797 -3.402088e-05 -0.0001283622 5.778250e-19
```

```
# Case 3 – unrestricted constant (baseline)
```

```
Jtest_trace_case3 <- ca.jo(data_coint,
                             type = "trace",
                             ecdet = "none",
                             K      = optimal_lag,
                             spec   = "transitory")
```

```
summary(Jtest_trace_case3)
```

```
##
## #####
## # Johansen-Procedure #
## #####
##
## Test type: trace statistic , with linear trend
##
## Eigenvalues (lambda):
## [1] 0.060997281 0.013822031 0.002359735
##
## Values of teststatistic and critical values of test:
##
##          test 10pct 5pct 1pct
## r <= 2 | 2.34 6.50 8.18 11.65
## r <= 1 | 16.15 15.66 17.95 23.52
## r = 0 | 78.58 28.71 31.52 37.22
##
## Eigenvectors, normalised to first column:
## (These are the cointegration relations)
##
##          HDK.l1  nom_ER.l1  EURIBOR.l1
## HDK.l1      1.0000000  1.0000000  1.0000000
## nom_ER.l1  -0.4054731 -4.5578506 -0.3237828
## EURIBOR.l1 -0.8273271 -0.3966417 0.4526235
##
## Weights W:
## (This is the loading matrix)
##
##          HDK.l1  nom_ER.l1  EURIBOR.l1
## HDK.d      -0.003803583  5.267385e-04 -5.442383e-03
## nom_ER.d    0.002870880  4.490554e-03 -3.092243e-05
## EURIBOR.d   0.002626308 -9.843917e-05 -1.504556e-04
```

```
# Case 2 – restricted trend
```

```
Jtest_trace_case2 <- ca.jo(data_coint,
                             type = "trace",
                             ecdet = "trend",
                             K      = optimal_lag,
                             spec   = "transitory")
```

```
summary(Jtest_trace_case2)
```

```
##
## #####
## # Johansen-Procedure #
## #####
##
## Test type: trace statistic , with linear trend in cointegration
##
## Eigenvalues (lambda):
## [1] 6.452785e-02 1.382735e-02 2.672488e-03 5.903337e-17
##
## Values of teststatistic and critical values of test:
##
##          test 10pct 5pct 1pct
## r <= 2 | 2.65 10.49 12.25 16.26
## r <= 1 | 16.47 22.76 25.32 30.45
## r = 0 | 82.64 39.06 42.44 48.45
##
## Eigenvectors, normalised to first column:
## (These are the cointegration relations)
##
##          HDK.l1  nom_ER.l1  EURIBOR.l1  trend.l1
## HDK.l1      1.0000000000  1.000000e+00  1.0000000000  1.0000000000
```

```

## nom_ER.l1 -0.8445688312 -4.658204e+00 -0.8951427599 1.0060438004
## EURIBOR.l1 -0.4758338193 -3.575300e-01 1.2959023069 -1.4709442303
## trend.l1 0.0002636002 2.284448e-05 0.0003805587 -0.0008750961
##
## Weights W:
## (This is the loading matrix)
##
## HDK.l1 nom_ER.l1 EURIBOR.l1 trend.l1
## HDK.d -0.003796201 0.0004895803 -4.265883e-03 1.745765e-16
## nom_ER.d 0.002611440 0.0044200828 -4.064469e-05 3.535539e-17
## EURIBOR.d 0.002354299 -0.0001008234 -1.263134e-04 2.078605e-17

## =====
## SECTION 5: VECM ESTIMATION
## =====
# One cointegrating vector confirmed by Johansen trace tests
# VECM with r=1 is appropriate
# Case 3 (unrestricted constant) used as baseline specification

# --- 5.1 Estimate VECM using cajorls ---
vecm_model <- cajorls(jtest_trace_case3, r = 1)

# Cointegrating vector (Long-run relationship)
vecm_model$beta

## ect1
## HDK.l1 1.0000000
## nom_ER.l1 -0.4054731
## EURIBOR.l1 -0.8273271

# Short-run dynamics and error correction terms for each equation
summary(vecm_model$rlm)

## Response HDK.d :
##
## Call:
## lm(formula = HDK.d ~ ect1 + constant + HDK.dl1 + nom_ER.dl1 +
## EURIBOR.dl1 + HDK.dl2 + nom_ER.dl2 + EURIBOR.dl2 - 1, data = data.mat)
##
## Residuals:
## Min 1Q Median 3Q Max
## -0.127488 -0.005317 0.000585 0.006146 0.054814
##
## Coefficients:
## Estimate Std. Error t value Pr(>|t|)
## ect1 -0.003804 0.003112 -1.222 0.222
## constant 0.009461 0.007776 1.217 0.224
## HDK.dl1 -0.015898 0.031909 -0.498 0.618
## nom_ER.dl1 -0.037990 0.066919 -0.568 0.570
## EURIBOR.dl1 0.473664 0.293324 1.615 0.107
## HDK.dl2 -0.009338 0.032835 -0.284 0.776
## nom_ER.dl2 -0.084170 0.064753 -1.300 0.194
## EURIBOR.dl2 -0.151509 0.292187 -0.519 0.604
##
## Residual standard error: 0.01172 on 984 degrees of freedom
## Multiple R-squared: 0.00665, Adjusted R-squared: -0.001426
## F-statistic: 0.8234 on 8 and 984 DF, p-value: 0.582
##
## Response nom_ER.d :
##
## Call:
## lm(formula = nom_ER.d ~ ect1 + constant + HDK.dl1 + nom_ER.dl1 +
## EURIBOR.dl1 + HDK.dl2 + nom_ER.dl2 + EURIBOR.dl2 - 1, data = data.mat)
##
## Residuals:
## Min 1Q Median 3Q Max
## -0.0243871 -0.0033592 -0.0000655 0.0032787 0.0281231
##
## Coefficients:
## Estimate Std. Error t value Pr(>|t|)
## ect1 0.002871 0.001495 1.920 0.0551 .
## constant -0.007212 0.003736 -1.931 0.0538 .
## HDK.dl1 0.120486 0.015329 7.860 1e-14 ***

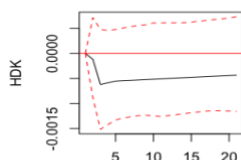
```

```
## nom_ER.dl1 -0.037547 0.032147 -1.168 0.2431
## EURIBOR.dl1 -0.128045 0.140910 -0.909 0.3637
## HDK.dl2 -0.020985 0.015773 -1.330 0.1837
## nom_ER.dl2 -0.028601 0.031107 -0.919 0.3581
## EURIBOR.dl2 -0.033266 0.140364 -0.237 0.8127
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.005632 on 984 degrees of freedom
## Multiple R-squared: 0.06756, Adjusted R-squared: 0.05998
## F-statistic: 8.912 on 8 and 984 DF, p-value: 7.62e-12
##
##
## Response EURIBOR.d :
##
## Call:
## lm(formula = EURIBOR.d ~ ect1 + constant + HDK.dl1 + nom_ER.dl1 +
## EURIBOR.dl1 + HDK.dl2 + nom_ER.dl2 + EURIBOR.dl2 - 1, data = data.mat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.0109409 -0.0003645  0.0000272  0.0005696  0.0124408
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## ect1          0.0026263  0.0003408   7.707 3.14e-14 ***
## constant     -0.0068063  0.0008516  -7.993 3.68e-15 ***
## HDK.dl1      -0.0083858  0.0034942  -2.400  0.01658 *
## nom_ER.dl1   -0.0121385  0.0073280  -1.656  0.09795 .
## EURIBOR.dl1  0.1233677  0.0321203   3.841 0.00013 ***
## HDK.dl2      -0.0016626  0.0035956  -0.462  0.64389
## nom_ER.dl2   -0.0082985  0.0070907  -1.170  0.24215
## EURIBOR.dl2  0.0963380  0.0319958   3.011 0.00267 **
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.001284 on 984 degrees of freedom
## Multiple R-squared: 0.1914, Adjusted R-squared: 0.1848
## F-statistic: 29.12 on 8 and 984 DF, p-value: < 2.2e-16

# --- 5.2 Convert to VAR representation for IRF and FEVD ---
vecm_var <- vec2var(Jtest_trace_case3, r = 1)

# --- 5.3 Impulse Response Functions ---
# Response of Euro STOXX 50 to shocks in nom_ER and EURIBOR
irf_nom_ER <- irf(vecm_var,
  impulse = "nom_ER",
  response = "HDK",
  n.ahead = 20,
  boot = TRUE,
  ci = 0.95)
plot(irf_nom_ER,
  main = "Response of Euro STOXX 50 to a shock in nom_ER")
```

Response of Euro STOXX 50 to a shock in nom_ER

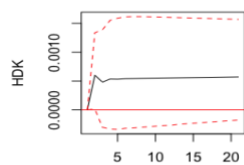


95 % Bootstrap CI, 100 runs

```
irf_EURIBOR <- irf(vecm_var,
  impulse = "EURIBOR",
  response = "HDK",
  n.ahead = 20,
  boot = TRUE,
  ci = 0.95)
```

```
plot(irf_EURIBOR,
     main = "Response of Euro STOXX 50 to a shock in EURIBOR")
```

Response of Euro STOXX 50 to a shock in EURIBOR

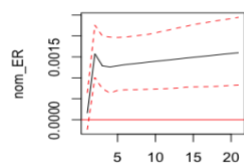


95 % Bootstrap CI, 100 runs

```
# Response of nom_ER and EURIBOR to a shock in Euro STOXX 50
```

```
irf_HDK_to_nom_ER <- irf(vecm_var,
                        impulse = "HDK",
                        response = "nom_ER",
                        n.ahead = 20,
                        boot = TRUE,
                        ci = 0.95)
plot(irf_HDK_to_nom_ER,
     main = "Response of nom_ER to a shock in Euro STOXX 50")
```

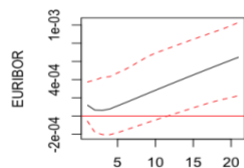
Response of nom_ER to a shock in Euro STOXX 50



95 % Bootstrap CI, 100 runs

```
irf_HDK_to_EURIBOR <- irf(vecm_var,
                        impulse = "HDK",
                        response = "EURIBOR",
                        n.ahead = 20,
                        boot = TRUE,
                        ci = 0.95)
plot(irf_HDK_to_EURIBOR,
     main = "Response of EURIBOR to a shock in Euro STOXX 50")
```

Response of EURIBOR to a shock in Euro STOXX 50



95 % Bootstrap CI, 100 runs

```
# --- 5.4 Forecast Error Variance Decomposition ---
```

```
# Shows what proportion of each variable's forecast variance is explained by shocks to the other variables
```

```
fevd_result <- fevd(vecm_var, n.ahead = 20)
fevd_result
```

```
## $HDK
##           HDK           nom_ER           EURIBOR
## [1,] 1.0000000 0.000000e+00 0.000000000
## [2,] 0.9986055 5.197366e-05 0.001342564
## [3,] 0.9974909 1.012924e-03 0.001496192
## [4,] 0.9968797 1.431262e-03 0.001689078
## [5,] 0.9965652 1.631305e-03 0.001803543
```

```

## [6,] 0.9963510 1.754329e-03 0.001894713
## [7,] 0.9961974 1.837986e-03 0.001964632
## [8,] 0.9960835 1.894833e-03 0.002021700
## [9,] 0.9959970 1.933329e-03 0.002069696
## [10,] 0.9959296 1.959088e-03 0.002111301
## [11,] 0.9958762 1.975638e-03 0.002148203
## [12,] 0.9958331 1.985318e-03 0.002181565
## [13,] 0.9957980 1.989756e-03 0.002212196
## [14,] 0.9957692 1.990121e-03 0.002240680
## [15,] 0.9957453 1.987272e-03 0.002267447
## [16,] 0.9957253 1.981854e-03 0.002292821
## [17,] 0.9957086 1.974364e-03 0.002317048
## [18,] 0.9956945 1.965186e-03 0.002340323
## [19,] 0.9956826 1.954625e-03 0.002362798
## [20,] 0.9956725 1.942924e-03 0.002384596
##
## $nom_ER
##          HDK      nom_ER      EURIBOR
## [1,] 0.0007776089 0.9992224 0.0000000000
## [2,] 0.0399170531 0.9596526 0.0004303099
## [3,] 0.0454634906 0.9539992 0.0005372792
## [4,] 0.0480599181 0.9512071 0.0007330077
## [5,] 0.0500436472 0.9491032 0.0008531700
## [6,] 0.0517819670 0.9472722 0.0009458787
## [7,] 0.0532974617 0.9456864 0.0010161221
## [8,] 0.0546675592 0.9442599 0.0010725053
## [9,] 0.0559446785 0.9429365 0.0011188649
## [10,] 0.0571575967 0.9416843 0.0011580843
## [11,] 0.0583243829 0.9404836 0.0011920219
## [12,] 0.0594572572 0.9393208 0.0012219620
## [13,] 0.0605647886 0.9381864 0.0012488031
## [14,] 0.0616530933 0.9370737 0.0012731943
## [15,] 0.0627266457 0.9359777 0.0012956146
## [16,] 0.0637887874 0.9348948 0.0013164254
## [17,] 0.0648420567 0.9338220 0.0013359039
## [18,] 0.0658884090 0.9327573 0.0013542663
## [19,] 0.0669293693 0.9316989 0.0013716840
## [20,] 0.0679661388 0.9306456 0.0013882942
##
## $EURIBOR
##          HDK      nom_ER      EURIBOR
## [1,] 0.008710202 0.019964527 0.9713253
## [2,] 0.005037987 0.013461570 0.9815004
## [3,] 0.003651595 0.009462113 0.9868863
## [4,] 0.003287441 0.007478438 0.9892341
## [5,] 0.003644685 0.006231720 0.9901236
## [6,] 0.004487253 0.005352620 0.9901601
## [7,] 0.005743264 0.004681585 0.9895752
## [8,] 0.007369171 0.004145400 0.9884854
## [9,] 0.009339525 0.003703514 0.9869570
## [10,] 0.011634293 0.003332386 0.9850333
## [11,] 0.014237769 0.003017421 0.9827448
## [12,] 0.017136559 0.002749074 0.9801144
## [13,] 0.020318622 0.002520795 0.9771606
## [14,] 0.023772678 0.002327898 0.9738994
## [15,] 0.027487882 0.002166900 0.9703452
## [16,] 0.031453626 0.002035116 0.9665113
## [17,] 0.035659428 0.001930411 0.9624102
## [18,] 0.040094870 0.001851031 0.9580541
## [19,] 0.044749567 0.001795494 0.9534549
## [20,] 0.049613159 0.001762517 0.9486243

```

```
# --- 5.5 Diagnostic Checks ---
```

```
# Serial correlation test
```

```
serial.test(vecm_var, type = "PT.adjusted")
```

```
##
```

```
## Portmanteau Test (adjusted)
```

```
##
```

```
## data: Residuals of VAR object vecm_var
```

```
## Chi-squared = 168.23, df = 120, p-value = 0.002441
```

```
# Test serial correlation across higher lag orders (K=4 to K=Lag_ADF)
```

```
# to check whether serial correlation persists or is resolved at higher lags
```



```

for(k in 4:lag_ADF) {
  Jtest_k <- ca.jo(data_coint, type="trace", ecdet="none", K=k, spec="transitory")
  vecm_k <- vec2var(Jtest_k, r=1)
  st <- serial.test(vecm_k, type="PT.adjusted")
  cat("Lag", k, "- p-value:", st$serial$p.value, "\n")
}

## Lag 4 - p-value: 0.002169828
## Lag 5 - p-value: 0.01643208
## Lag 6 - p-value: 0.004909947
## Lag 7 - p-value: 0.02416527
## Lag 8 - p-value: 0.01794908
## Lag 9 - p-value: 0.005882953
## Lag 10 - p-value: 0.002836409
## Lag 11 - p-value: 0.01760976
## Lag 12 - p-value: 0.004125544
## Lag 13 - p-value: 0.01043145
## Lag 14 - p-value: 0.001218585
## Lag 15 - p-value: 0.0001664321
## Lag 16 - p-value: 1.215587e-05

## Warning in pchisq(STATISTIC, df = PARAMETER): NaNs produced
## Warning in pchisq(STATISTIC, df = PARAMETER): NaNs produced

## Lag 17 - p-value: NaN

## Warning in pchisq(STATISTIC, df = PARAMETER): NaNs produced
## Warning in pchisq(STATISTIC, df = PARAMETER): NaNs produced

## Lag 18 - p-value: NaN

## Warning in pchisq(STATISTIC, df = PARAMETER): NaNs produced
## Warning in pchisq(STATISTIC, df = PARAMETER): NaNs produced

## Lag 19 - p-value: NaN

## Warning in pchisq(STATISTIC, df = PARAMETER): NaNs produced
## Warning in pchisq(STATISTIC, df = PARAMETER): NaNs produced

## Lag 20 - p-value: NaN

## Warning in pchisq(STATISTIC, df = PARAMETER): NaNs produced
## Warning in pchisq(STATISTIC, df = PARAMETER): NaNs produced

## Lag 21 - p-value: NaN

# Normality test
normality.test(vecm_var, multivariate.only = TRUE)

## $JB
##
## JB-Test (multivariate)
##
## data: Residuals of VAR object vecm_var
## Chi-squared = 19147, df = 6, p-value < 2.2e-16
##
##
## $Skewness
##
## Skewness only (multivariate)
##
## data: Residuals of VAR object vecm_var
## Chi-squared = 411.86, df = 3, p-value < 2.2e-16
##
##
## $Kurtosis
##
## Kurtosis only (multivariate)
##
## data: Residuals of VAR object vecm_var
## Chi-squared = 18735, df = 3, p-value < 2.2e-16

```

```

# ARCH effects test
arch.test(vecm_var, lags.multi = 3, multivariate.only = TRUE)

##
## ARCH (multivariate)
##
## data: Residuals of VAR object vecm_var
## Chi-squared = 298.72, df = 108, p-value < 2.2e-16

## =====
## SECTION 6: EXPORT TO CSV
## =====
# Export for Robinson (1995) fractional cointegration test in Stata

# Also save the ECT residuals (cointegrating residuals)
# These are the residuals from the cointegrating relationship:  $\beta'y_{t-1}$ 
ect_residuals <- as.numeric(data_coint %*% vecm_model$beta)
ect_df <- data.frame(
  date = index(ln_merged),
  resVECM = ect_residuals,
  HDK = as.numeric(ln_merged$HDK),
  nom_ER = as.numeric(ln_merged$nom_ER),
  EURIBOR = as.numeric(ln_merged$EURIBOR)
)
write.csv(ect_df, "fractional_cointegration_data.csv", row.names = FALSE)

```

8.2 Stata Do-File

```

-----
name: <unnamed>
log: /Users/hidde/Library/CloudStorage/OneDrive-UniversityofBristol
> /Documents/Applied Financial Econometrics/Coursework/fractional_cointegra
> tion_results.log
log type: text
opened on: 17 Apr 2026, 20:51:19

```

```

.
. * Set working directory
. cd "/Users/hidde/Library/CloudStorage/OneDrive-UniversityofBristol/Docume
> nts/Applied Financial Econometrics/Coursework/"
/Users/hidde/Library/CloudStorage/OneDrive-UniversityofBristol/Documents/Ap
> plied Financial Econometrics/Coursework

```

```

.
. * Import data exported from R
. import delimited "fractional_cointegration_data.csv", clear
(encoding automatically selected: ISO-8859-1)
(5 vars, 995 obs)

```

```

.
. * Check data loaded correctly
. describe

```

Contains data

```

Observations:      995
Variables:         5

```

Variable name	Storage type	Display format	Value label	Variable label
date	str10	%10s		
resvecm	float	%9.0g		resVECM
hdk	float	%9.0g		HDK
nom_er	float	%9.0g		nom_ER
euribor	float	%9.0g		EURIBOR

Sorted by:

Note: Dataset has changed since last saved.

```

. summarize

```

Variable	Obs	Mean	Std. dev.	Min	Max
date	995				
resvecm	995				
hdk	995				
nom_er	995				
euribor	995				

date		0			
resvecm		995	2.476048	.1312858	2.164311 2.707718
hdk		995	2.332693	.0889162	2.110213 2.523326
nom_er		995	.1231984	.0413728	.0383039 .223944
euribor		995	-.2336553	.1312997	-.332 .076

```
.
. gen t = _n
```

```
. tsset t
```

```
Time variable: t, 1 to 995
Delta: 1 unit
```

```
.
. roblpr resvecm
```

Robinson estimates of fractional differencing parameter for resvecm

Power	Ords	Est d	Std Err	t(H0: d=0)	P> t
.9	499	.8734479	.0284389	30.7131	0.000

```
. * d < 1 but > 0.5 suggests possible long memory / fractional cointegratio
> n
```

```
.
. log close
  name: <unnamed>
  log: /Users/hidde/Library/CloudStorage/OneDrive-UniversityofBristol
> /Documents/Applied Financial Econometrics/Coursework/fractional_cointegra
> tion_results.log
  log type: text
closed on: 17 Apr 2026, 20:51:24
```